

Motor Datasheet

Operating Point: Continuous | Number of Turns per Tooth: 22

Geometry

Parameter	Value	Unit
Machine Outer Diameter	55.0	mm
Machine Inner Diameter	20.0	mm
Machine Thickness	24.2	mm
Airgap	1.0	mm

Mass Properties (active elements only)

Parameter	Value	Unit
Machine Mass	272.3	g
Rotor Mass	61.1	g
Stator Mass	211.2	g
Rotor Moment of Inertia	248.7	g cm2

Electrical Properties

Parameter	Value	Unit
Supply Current (RMS)	5.8	A
Phase Resistance	84.4	mOhm
D-Axis Inductance (Ld)	257.6	μH
Q-Axis Inductance (Lq)	255.4	μH
PM Flux Linkage	2.2	mWb
Frequency at operating point (electrical)	237.5	Hz
Winding Configuration	wye (star)	-

Performance

Parameter	Value	Unit
Torque (Continuous)	141.2	mNm
Speed	2850.0	rpm
BackEMF Constant (Ke)	1.2	mV/rpm
Motor Constant (Km, only resistive losses)	46.0	mNm/sqrt(W)
Torque Constant (Kt, no-load)	16.2	mNm/A_peak
Saturation	0.0	%

Losses

Parameter	Value	Unit
Total Losses	9.4	W
Resistive Losses	8.5	W

Configuration

Parameter	Value	Unit
Pole Pairs	5	-
Number of Turns per Tooth	22	-
Number of Phase Turns	88	-